

Design and Implementation of T-Logic Based Arithmetic Circuits in QCA Technology

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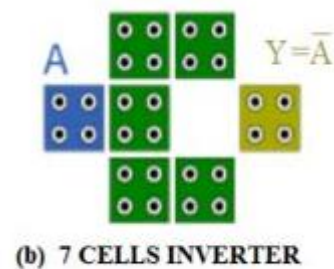
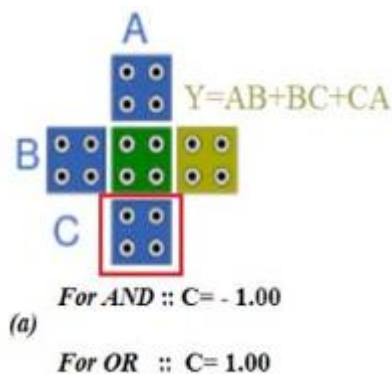
Branch : ECE

Course :VLSI

Introduction :

Quantum-dot Cellular Automata (QCA) offers an alternative to CMOS by addressing its scaling limitations and enabling room-temperature operation. QCA circuits are built using basic elements such as the Majority Voter (MV), inverter, and binary wire. The MV gate consists of 5 cells and accepts three inputs (A, B, C). By fixing the control input C to either 0 or 1, the MV can function as an AND or OR gate. Its output is given by the expression

$$Y = AB + BC + CA.$$



Full Adder :

According to this truth table, mathematical logics of sum and carry for full adder comes out to be:

$$\text{Sum} = A \oplus B \oplus \text{Cin}$$

$$\text{Cout} = AB + BC\text{in} + AC\text{in}$$

As discussed earlier, the Cout logic can be re-written as majority of A, B and C.
i.e. $Cout = M(A, B, Cin)$

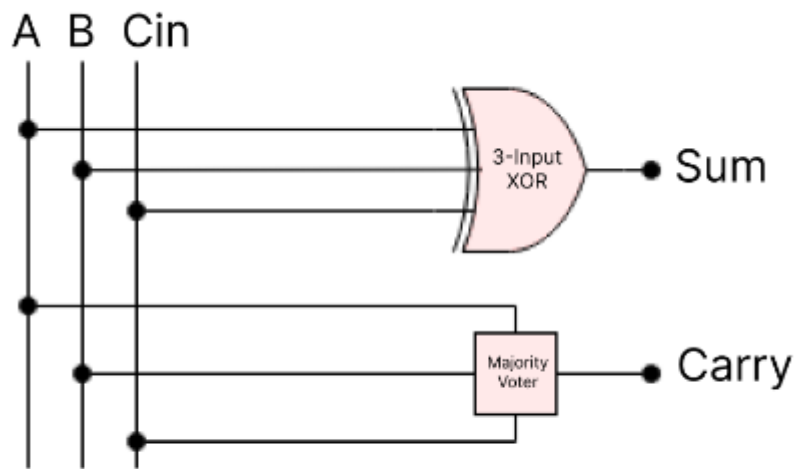
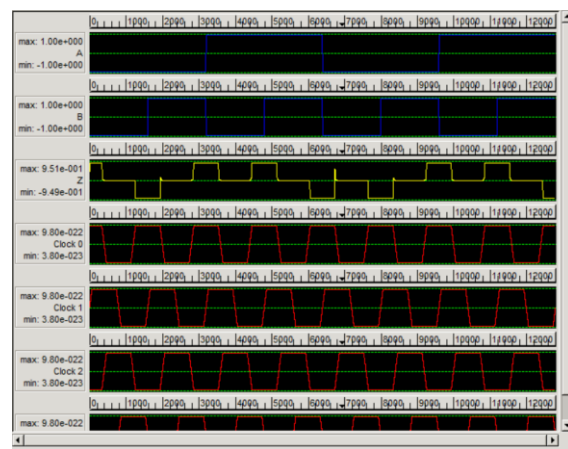
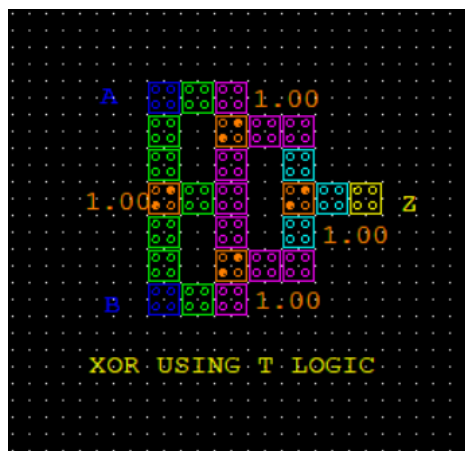
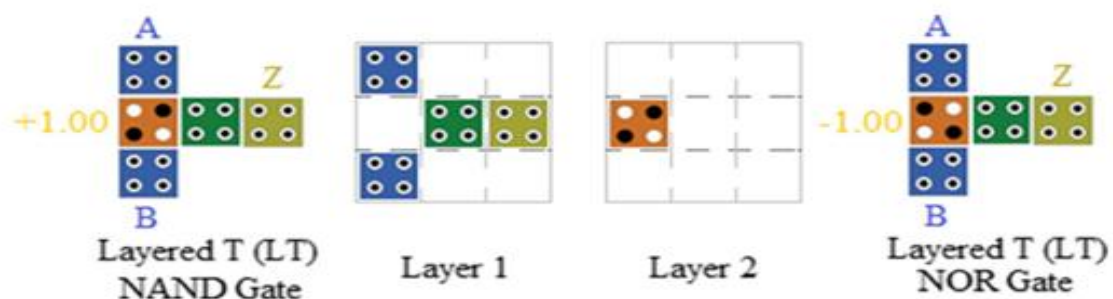


Fig (4) : Schematic of Full Adder as per equations.

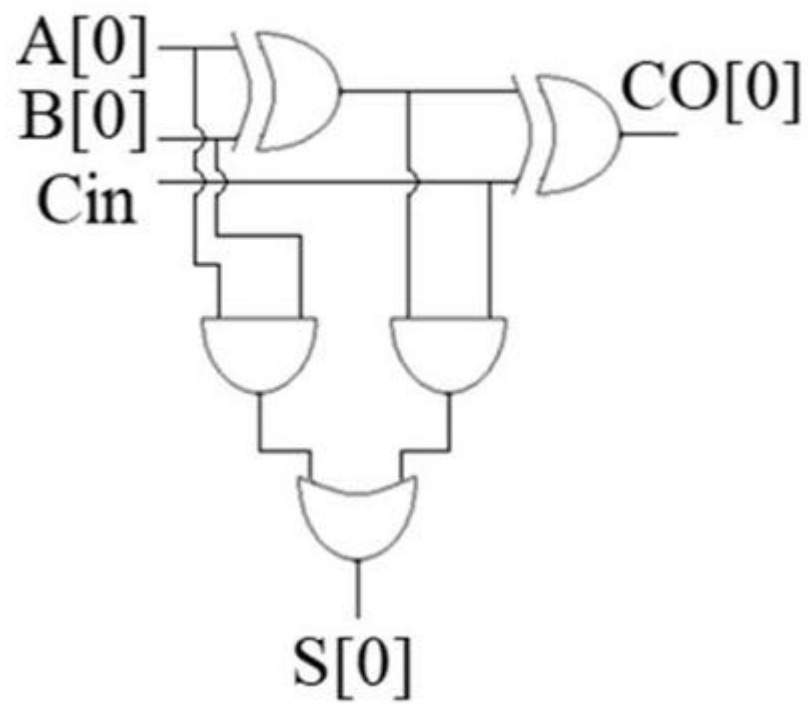
XOR gate using T-logic :-



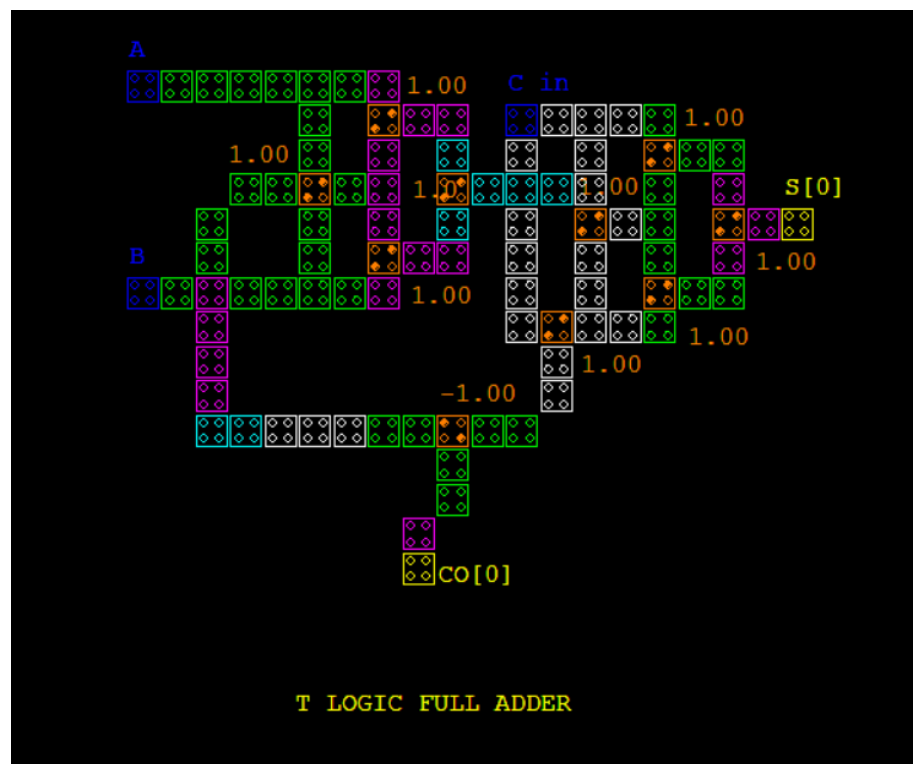
Layered T logic :-



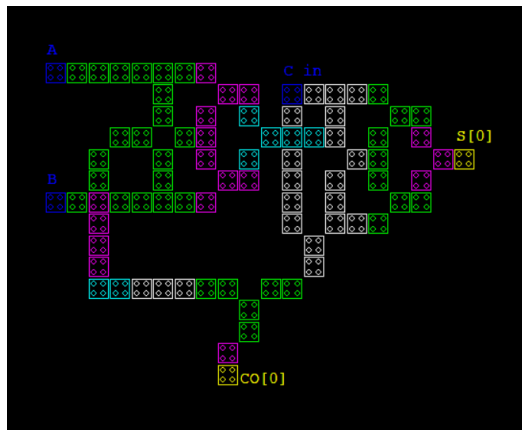
Full Adder using T-logic :-



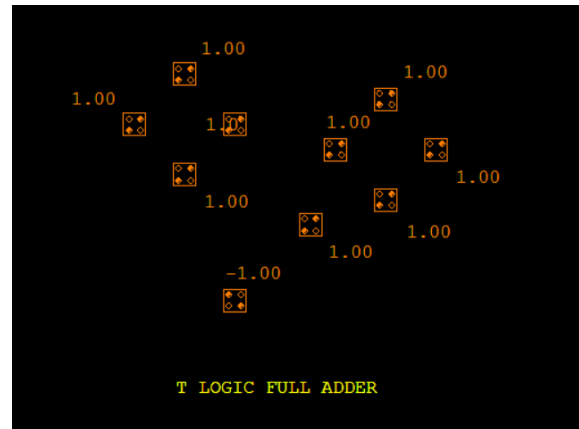
QCA Design :-



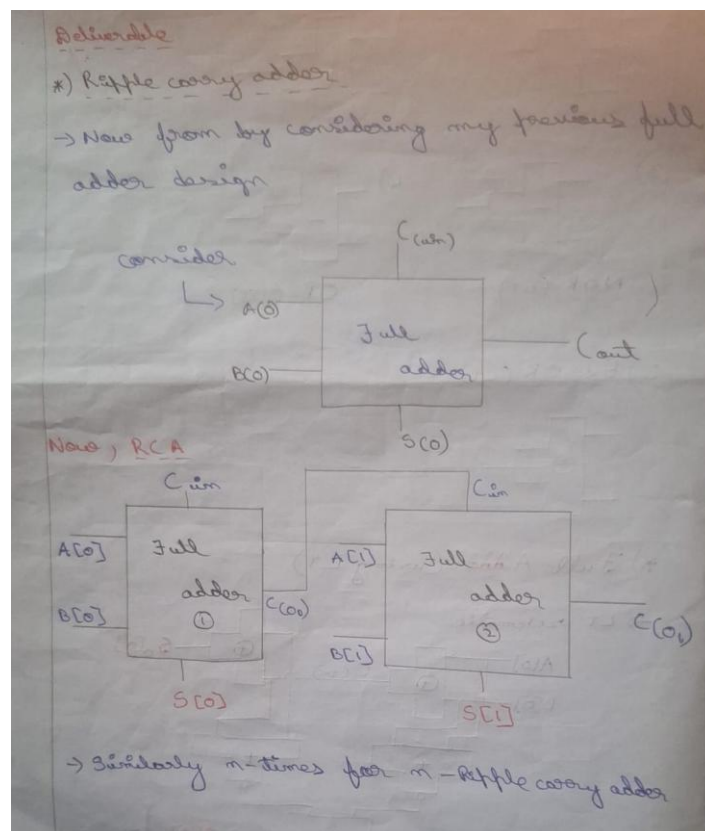
Layer 1:-



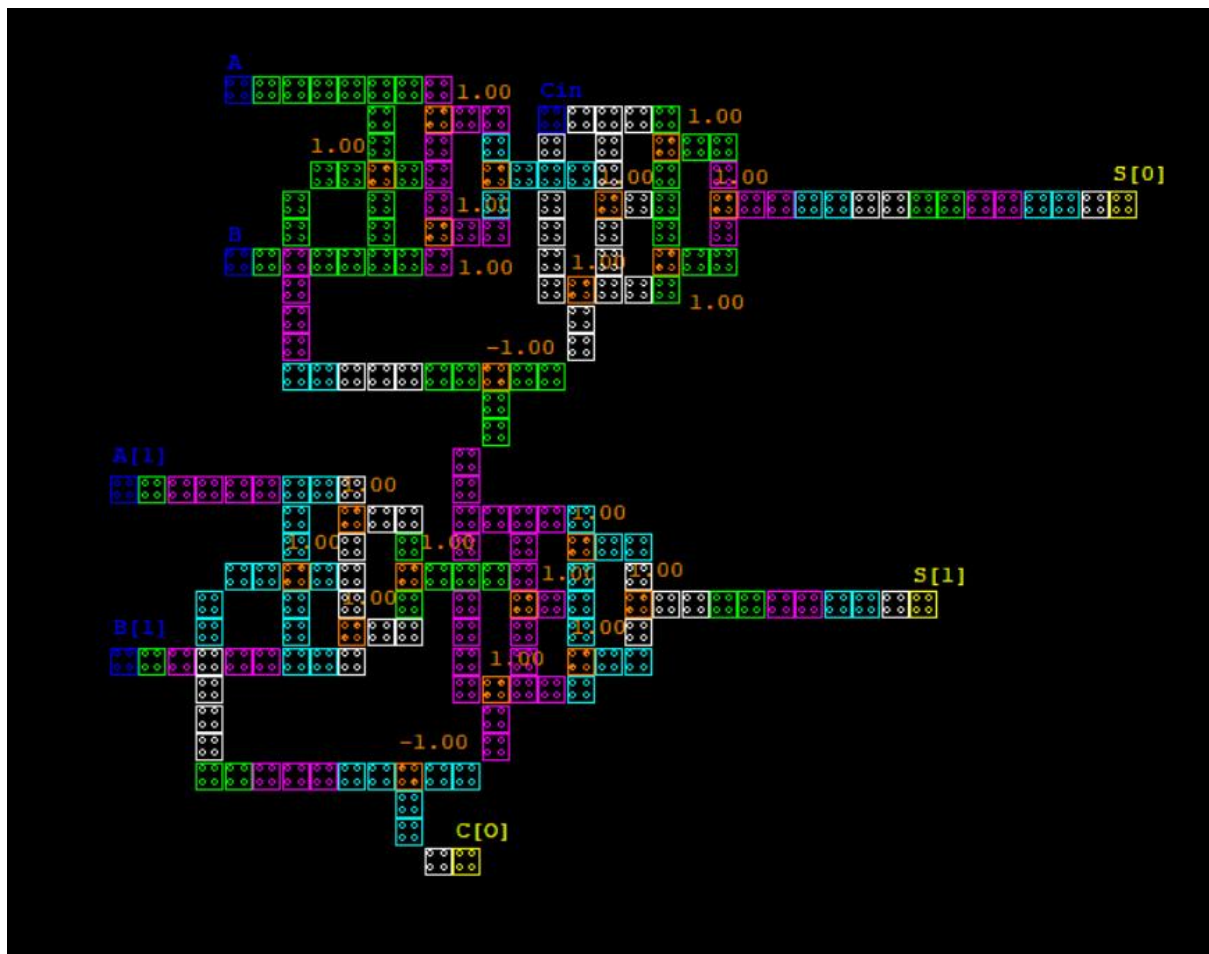
Layer 2:-



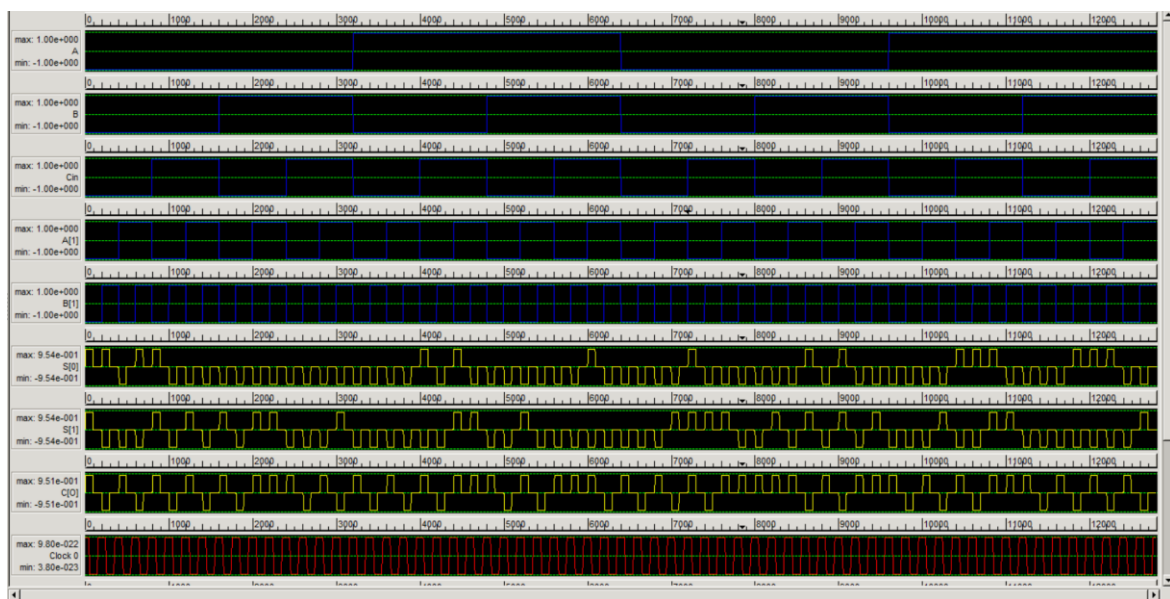
Ripple Carry Adder :-



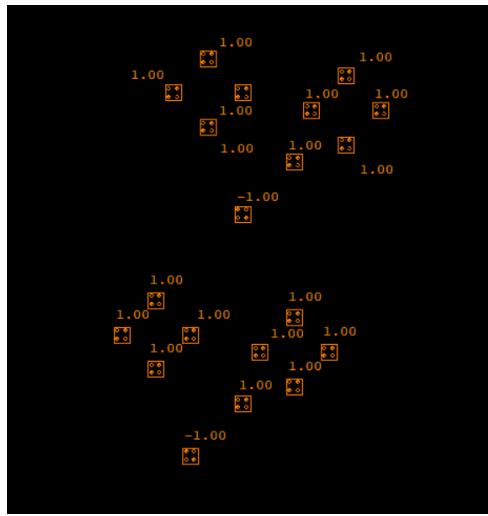
2-Bit RCA using T-logic :-



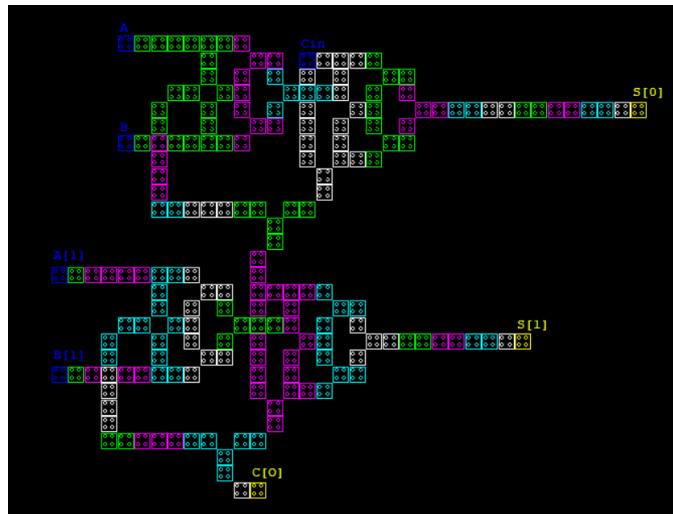
OUTPUT



Layer 2:-

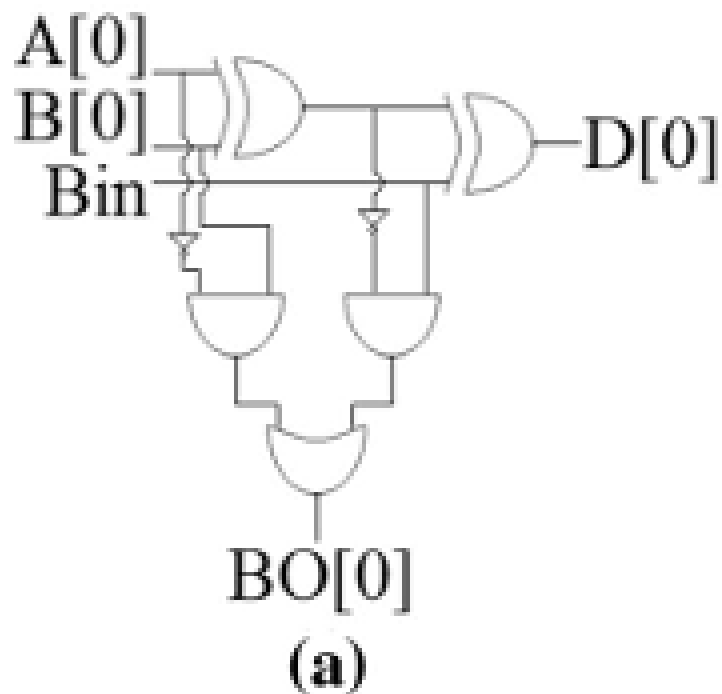


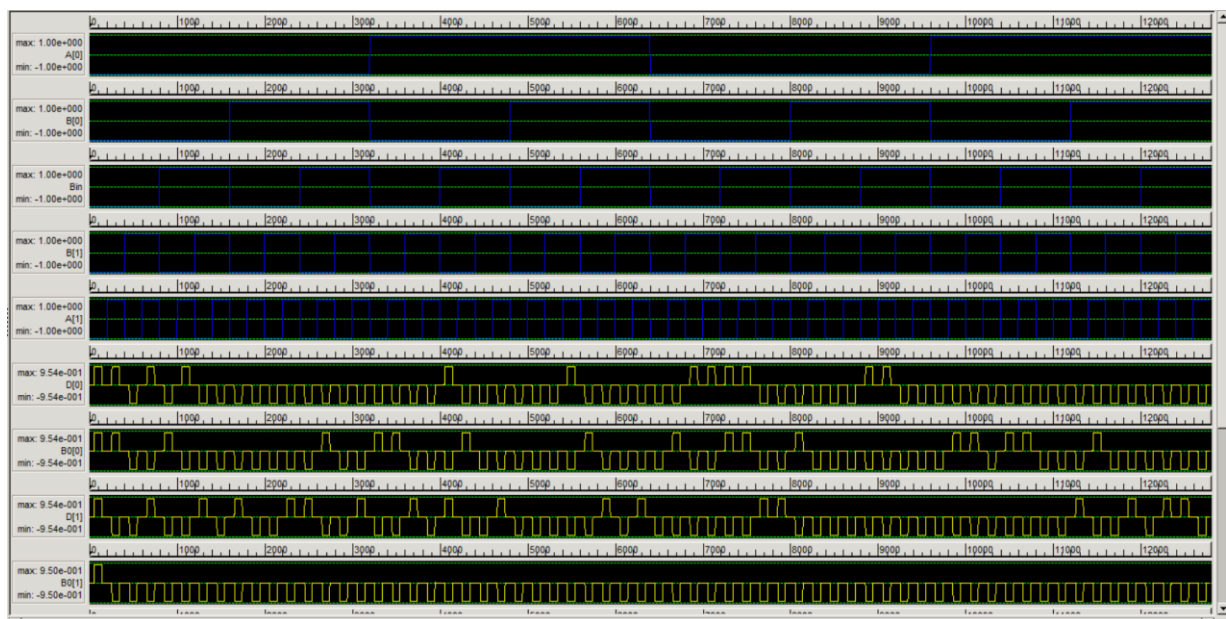
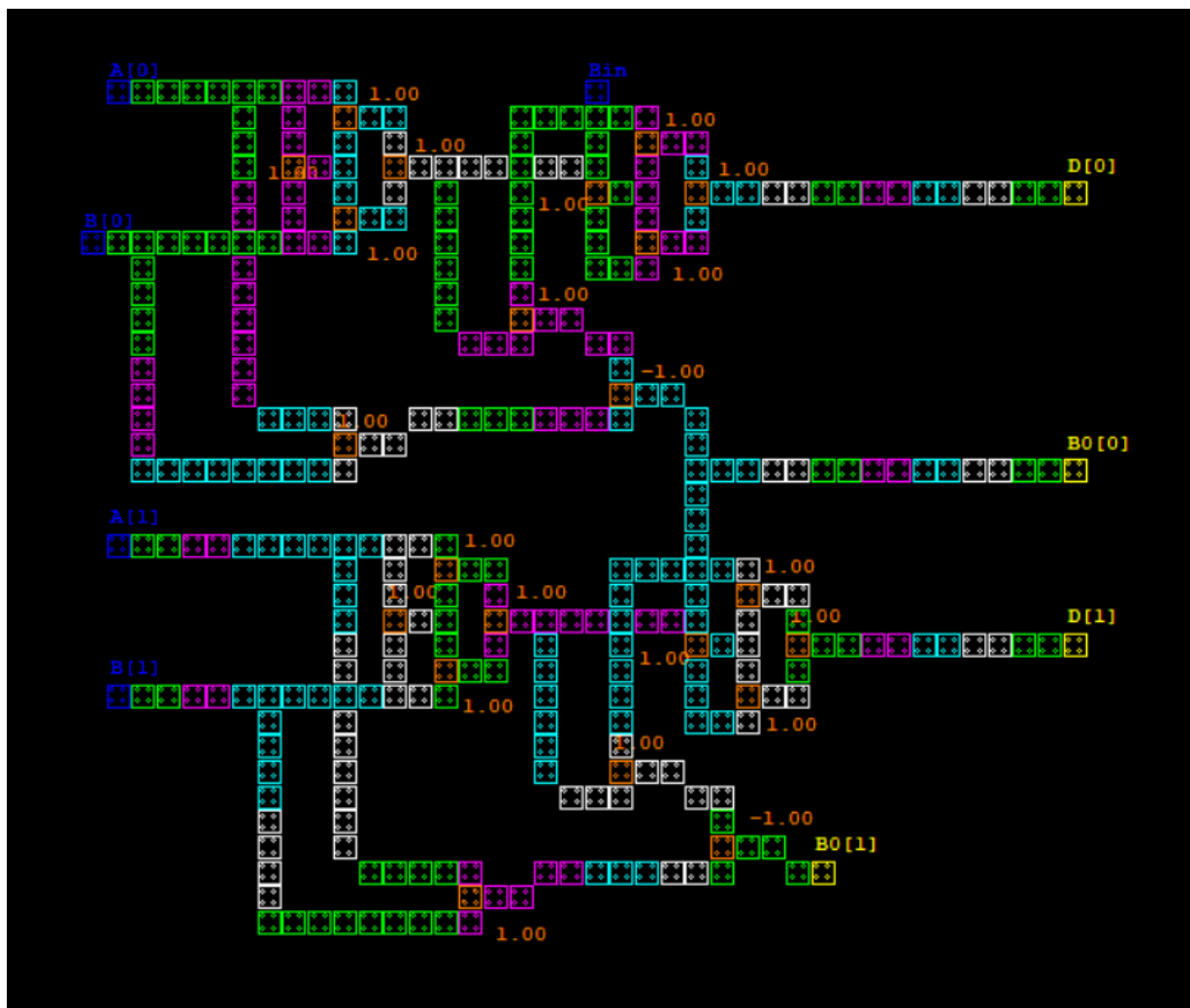
Layer 1:-



2-Bit RCS using T-logic :-

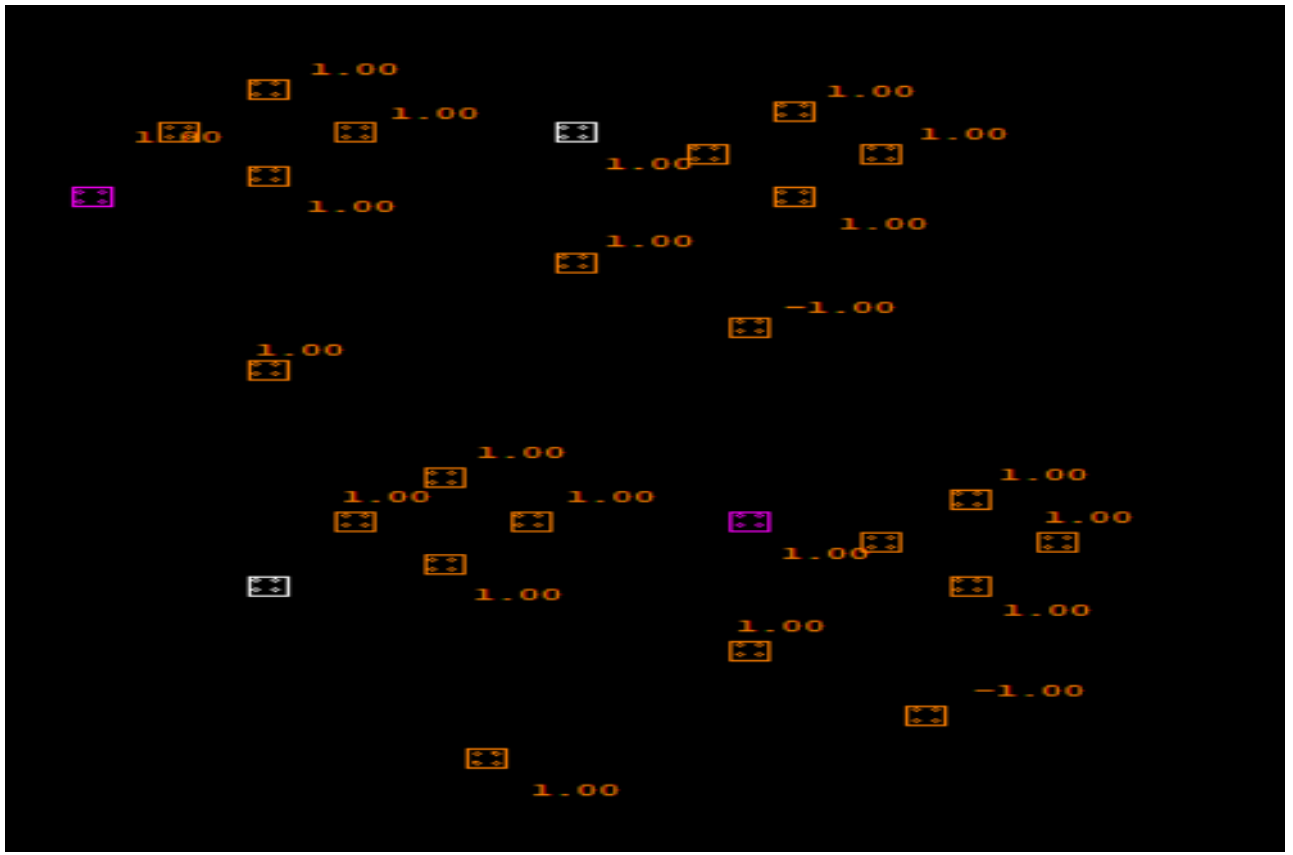
Similarly by using T-logic we build the 2bit ripple carry subtractor



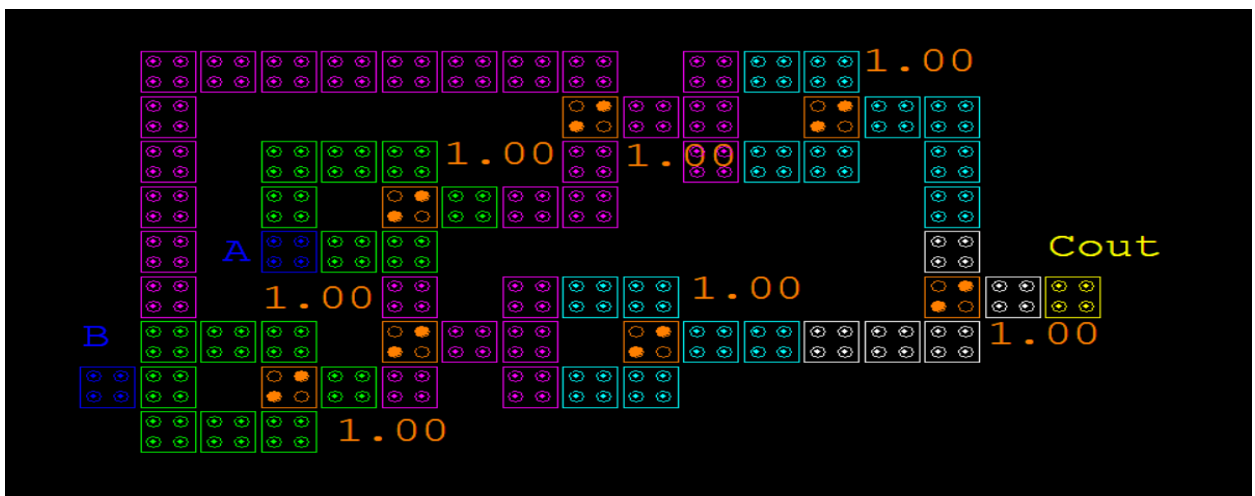


LAYERS :-

Here We use Both T layer , Multilayered structures



1-BIT COMPARATOR USING T-LOGIC :-



Layer 2:-



Output

